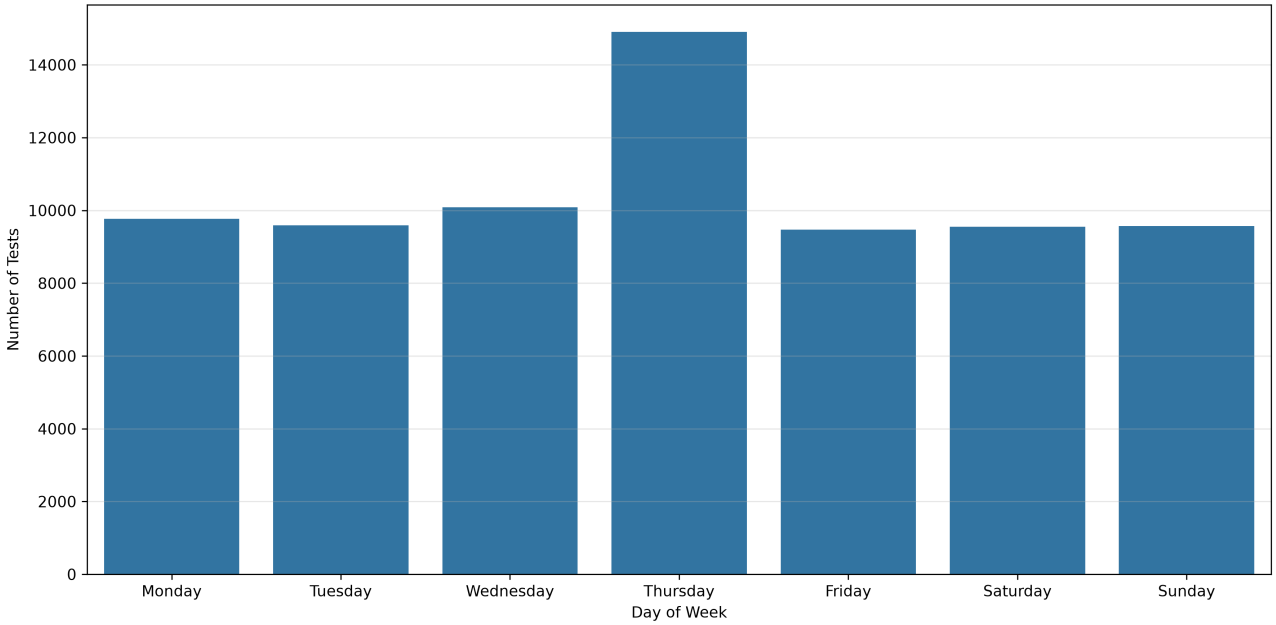
Lab Test Volume by Day

This following chart shows the distribution of laboratory tests across weekdays to identify variations in daily workload and determine whether certain days experience higher testing demand.



Key insights

- Thursday has the highest workload, with 14,000+ tests, considerably higher than the other weekdays.
- Test volume on the remaining weekdays remains relatively consistent at around 9,000 tests per day.
- The Thursday peak represents roughly 5,000+ additional tests compared with the typical daily workload, making it the main day requiring attention.

Potential bottleneck

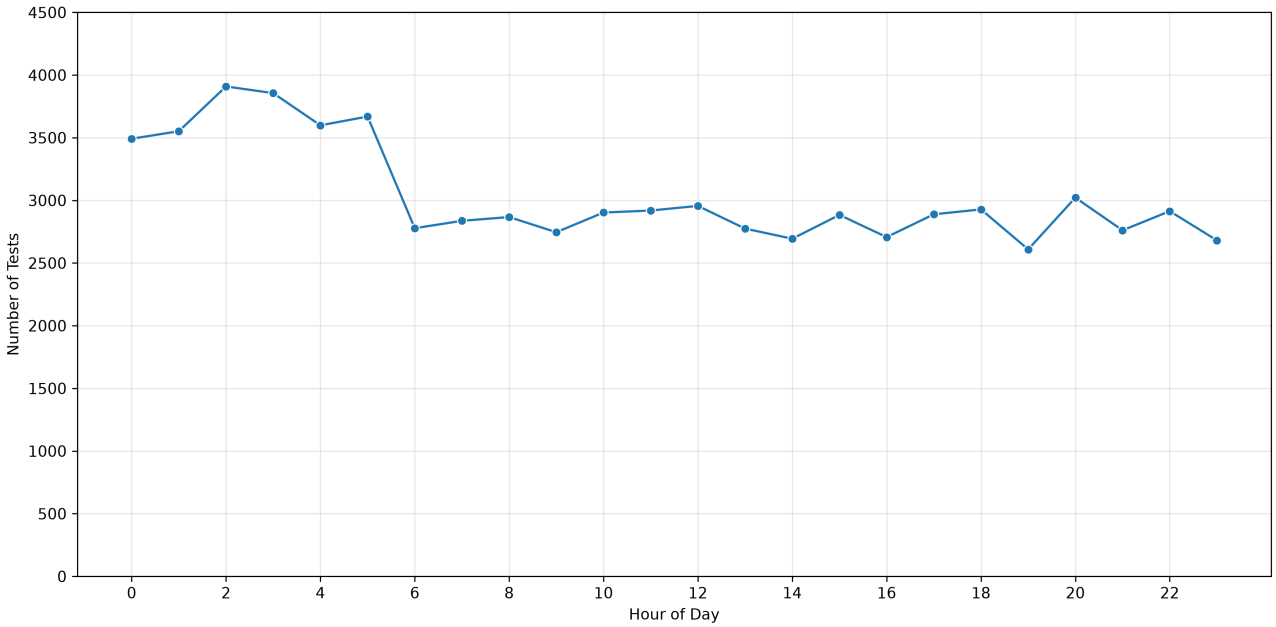
The concentration of additional workload on Thursday could put greater pressure on laboratory staff, analyzers, sample processing, and turnaround times if capacity is not adjusted accordingly.

Suggested action

Review staffing, analyzer utilization, and sample-processing capacity on Thursdays. If the higher demand is recurring, consider allocating additional resources or redistributing workload where operationally feasible.

Hourly Laboratory Test Volume

This chart shows the distribution of laboratory tests across different hours in UTC of the day to identify peak testing periods and understand when workload is concentrated.



Key insights

- Test volume remains relatively high between 00:00 and 5:00 UTC, indicating the busiest period of laboratory activity.
- After 5:00, test volume declines, suggesting reduced workload throughtout the day.
- The concentration of testing from midnight through the early morning indicates that this is the primary period of operational demand.

Potential bottleneck

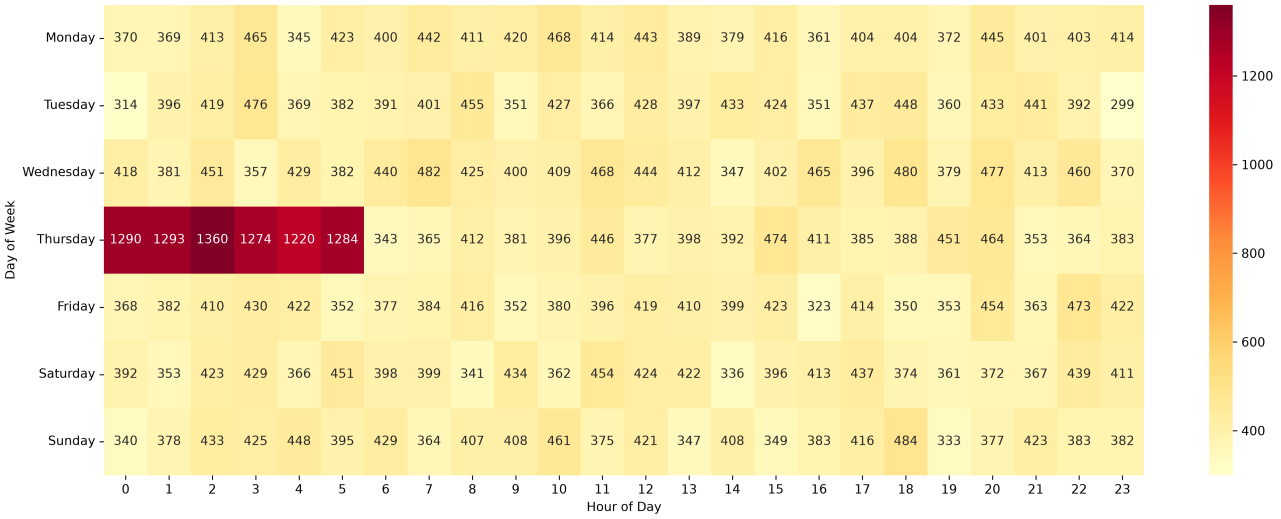
The sustained high workload between 00:00 and 5:00 UTC could create pressure on staff, analyzers, and sample-processing capacity, particularly if resources are not aligned with this demand.

Suggested action

Consider aligning staffing schedules, analyzer availability, and sample-processing resources with the 00:00 and 5:00 UTC peak to maintain smooth workflow and minimize the risk of delays.

Workload by Day and Hour

This heatmap combines day-of-week and hourly test volumes to identify when laboratory workload is most concentrated and whether specific days experience stronger peak periods.



Key insights

- The highest workload is concentrated on Thursday between 00:00 and 5:00 UTC, where the heatmap shows the darkest intensity.
- The overlap of these two peaks indicates that Thursday from midnight to early morning is the most demanding operational window.

Potential bottleneck

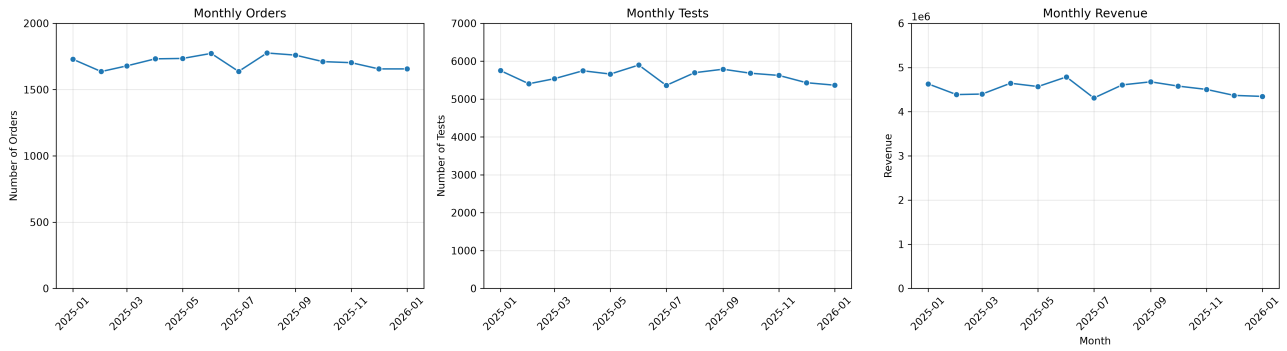
The combination of high daily volume and peak operational demand on Thursday may place additional pressure on staff, analyzers, and sample-processing capacity during this specific window.

Suggested action

Prioritize staffing, analyzer availability, and workflow capacity on Thursday afternoons, particularly between 00:00 and 5:00, and monitor TAT during this period to determine whether the increased workload is translating into delays.

Monthly Laboratory Demand Trend

This line chart compares monthly orders, tests, and revenue to understand overall laboratory activity, identify periods of changing demand, and determine whether operational and financial metrics move together.

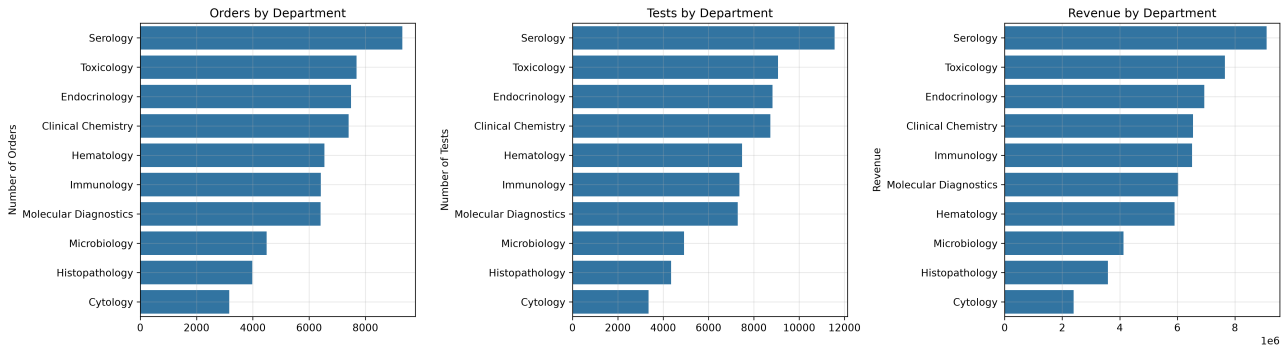


Key insights

- Orders, tests, and revenue generally follow the same monthly pattern, indicating that changes in patient volume are accompanied by changes in laboratory activity and revenue.
- July 2025 shows the only notable decline across all four metrics, suggesting a temporary reduction in overall laboratory activity.

Department demand trend

This chart compares tests, orders, and revenue across laboratory departments to identify where workload is concentrated and which departments contribute most to overall laboratory activity and revenue.



Key insights

- Serology has the highest workload, recording the largest number of tests and orders among the departments.
- The department also generates the highest revenue, showing that its high testing volume translates into a substantial financial contribution.
- The alignment between tests, orders, and revenue suggests that departments handling greater testing demand are also contributing more to overall revenue.

Potential bottleneck

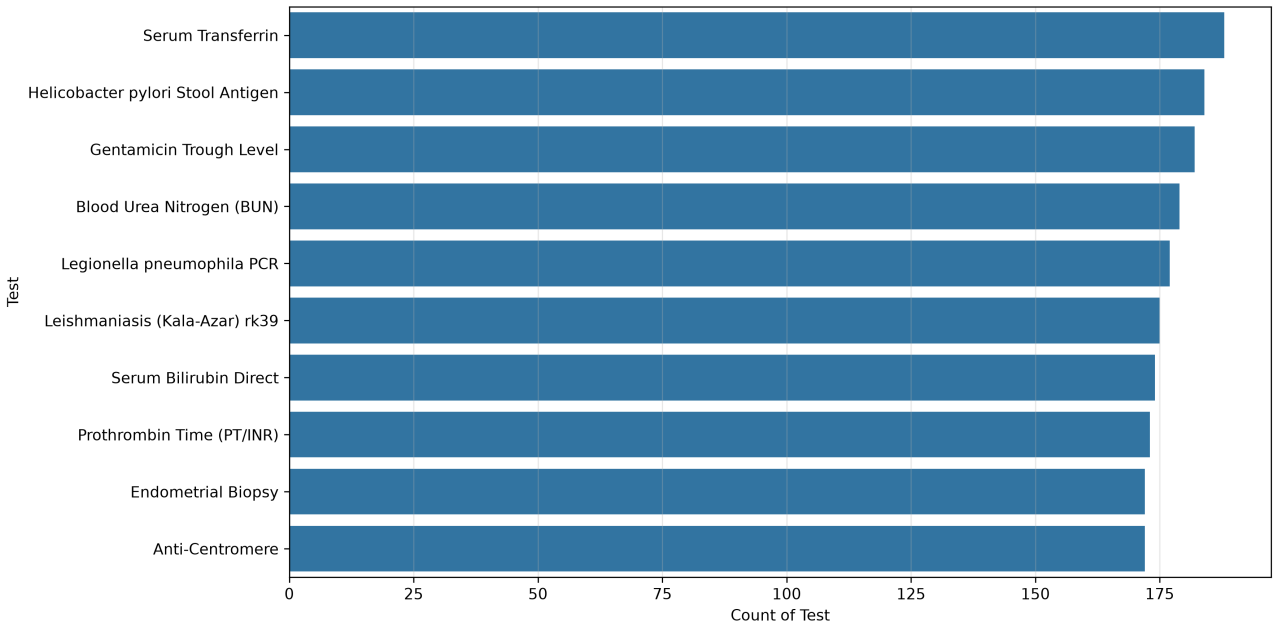
The concentration of workload in Serology may create a potential capacity pressure if staffing, analyzers, or other resources are not aligned with its higher testing demand.

Suggested action

Review Serology's staffing, analyzer utilization, and processing capacity against its workload. Maintaining sufficient capacity in this department can help manage demand while protecting turnaround times and revenue contribution.

Top 10 most frequently performed tests

This chart ranks the 10 most frequently performed laboratory tests by test volume to identify which tests contribute most to overall testing demand.



Key insights

- Serum Transferrin is the most frequently performed test, indicating the highest testing demand among the top 10 tests.
- The remaining tests show varying levels of demand, with the chart highlighting a clear ranking of the most frequently requested tests.
- The concentration of demand among these tests can help identify which tests require consistent reagent availability, analyzer capacity, and workflow planning.

Potential bottleneck

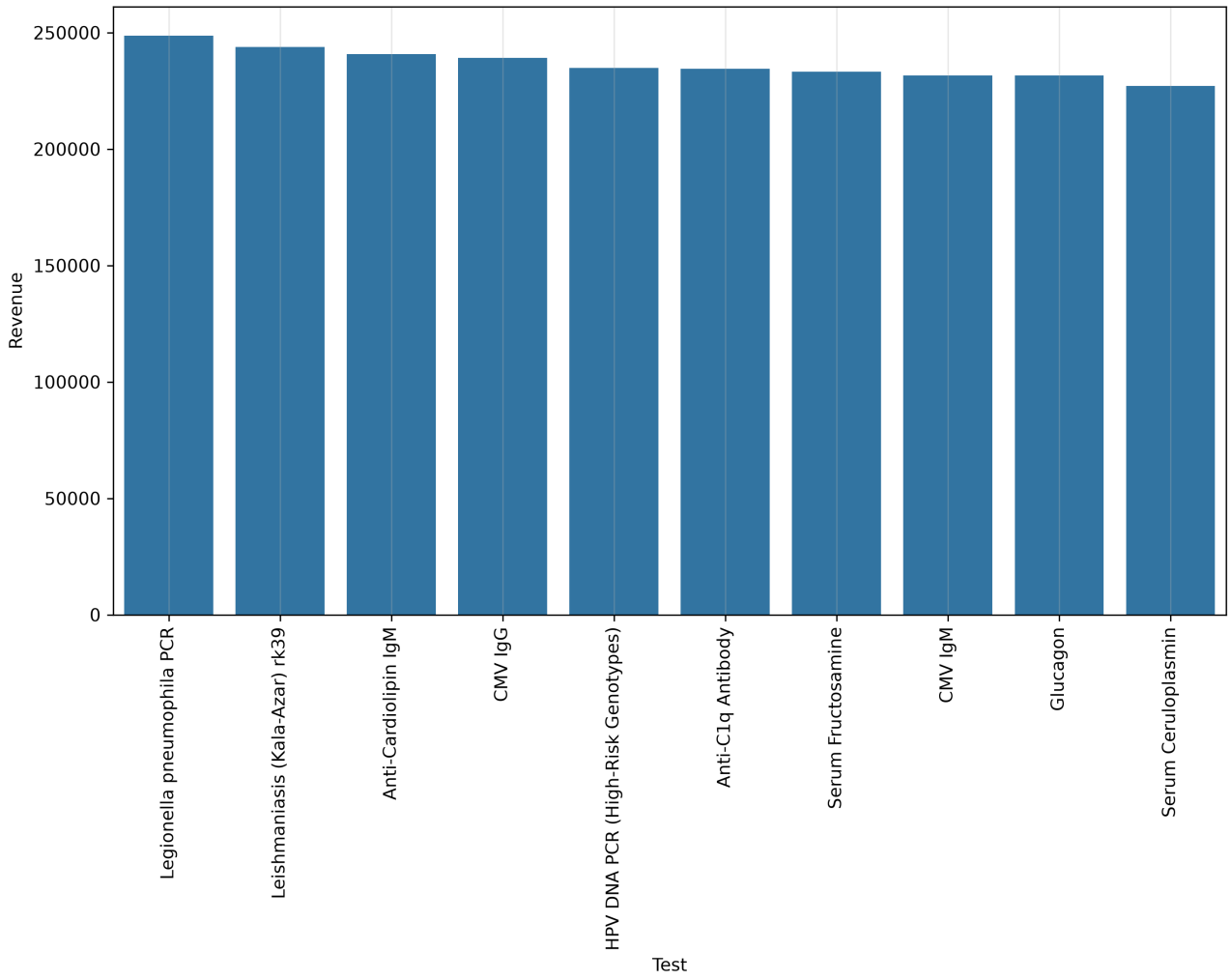
High demand for Serum Transferrin could create a potential capacity or supply bottleneck if reagent availability, analyzer capacity, or processing resources are not aligned with its testing volume.

Suggested action

Monitor Serum Transferrin demand, reagent consumption, and analyzer utilization to ensure sufficient capacity and avoid interruptions in testing.

Top 10 tests by revenue

This chart ranks the 10 laboratory tests generating the highest revenue to identify which tests contribute most to the laboratory's financial performance.



Key insights

- Legionella pneumophilia PCR is the highest revenue-generating test, contributing approximately 250,000 in revenue.
- Interestingly, the most frequently performed test, Serum Transferrin, does not generate the highest revenue, showing that testing volume and financial contribution do not necessarily follow the same pattern.
- This indicates that test value/pricing can have a stronger impact on revenue than testing frequency alone.

Potential bottleneck

High-volume tests may consume substantial reagents, analyzer capacity, and staff time without being the largest contributors to revenue. Focusing only on test volume could therefore overlook financially important tests.

Suggested action

Evaluate tests using both volume and revenue contribution. High-revenue tests such as Legionella pneumophilia PCR should be monitored for reagent availability and processing capacity, while high-volume, lower-revenue tests should be assessed for their resource utilization and operational efficiency.